

五 定积分

1 定积分的概念与性质

定义 设 $f(x)$ 在 (a, b) 上有定义, 用分点 $a = x_1 < x_2 < \dots < x_i < \dots < x_{n+1} = b$

将 $[a, b]$ 分割为 n 个小区间 $[x_i, x_{i+1}]$, 记 $\Delta x_i = x_{i+1} - x_i$,

$\lambda = \max_{1 \leq i \leq n} \{\Delta x_i\}$, 任取 $\xi_i \in [x_i, x_{i+1}]$, $i = 1, 2, \dots, n$,

若 $\lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$ 存在且其值与 x_i 和 ξ_i 的选取无关,

则称 $f(x)$ 在 $[a, b]$ 上可积, 称该极限值为 $f(x)$ 在 $[a, b]$ 上的定积分, 记为 $\int_a^b f(x) dx = \lim_{\lambda \rightarrow 0} \sum_{i=1}^n f(\xi_i) \Delta x_i$.

定理 若 $f(x)$ 在 $[a, b]$ 上可积, 则 $f(x)$ 在 $[a, b]$ 上有界.

定理 若 $f(x)$ 在 $[a, b]$ 上连续, 则 $f(x)$ 在 $[a, b]$ 上可积.

定理 若 $f(x)$ 在 $[a, b]$ 上有界且 $f(x)$ 在 (a, b) 只有第一类间断点, 则 $f(x)$ 在 $[a, b]$ 上可积.

性质 1° $\int_a^b f(x) dx = \int_a^b f(t) dt$

2° $\int_a^b f(x) dx = -\int_b^a f(x) dx$ (有向性)

3° $\int_a^b 1 dx = b - a$

4° $\int_a^b (\lambda f(x) + \mu g(x)) dx = \lambda \int_a^b f(x) dx + \mu \int_a^b g(x) dx$

5° $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$,

$$6^\circ x \in [a, b], f(x) \leq g(x), \int_a^b f(x) dx \leq \int_a^b g(x) dx$$

$$7^\circ \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \quad (a < b)$$

$$8^\circ x \in [a, b], m \leq f(x) \leq M, m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$9^\circ f(x) \in C(a, b), g(x) \in C(a, b) \text{ 且在 } (a, b) \text{ 上不变号.}$$

$$\text{则 } \exists \xi \in [a, b], \text{ 使 } \int_a^b f(x) dx = f(\xi) \int_a^b g(x) dx. \text{ (积分中值定理)}$$

$$10^\circ f(x) = -f(-x), x \in [-a, a], \int_{-a}^a f(x) dx = 0$$

$$f(x) = f(-x), x \in [-a, a], \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx = 2 \int_{-a}^0 f(x) dx$$

$$11^\circ \text{ 设 } f(x) \text{ 在 } \mathbb{R} \text{ 上以 } T \text{ 为周期, 则 } \int_0^{a+T} f(x) dx = \int_0^T f(x) dx.$$

$$\text{例: 设 } f(x), g(x) \in R[a, b],$$

$$\text{证明: } \left(\int_a^b f(x) g(x) dx \right)^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx \quad (*)$$

$$\text{证. 令 } h(t) = \int_a^b [t f(x) - g(x)]^2 dx$$

$$= t^2 \int_a^b f^2(x) dx - 2t \int_a^b f(x) g(x) dx + \int_a^b g^2(x) dx$$

$$\text{易知 } h(t) \geq 0.$$

$$1^\circ \text{ 若 } \int_a^b f(x) dx = 0, \text{ 则 } \int_a^b f(x) g(x) dx = 0$$

$$2^\circ \text{ 若 } \int_a^b f(x) dx \neq 0, \text{ 则 } \Delta = \left(-2 \int_a^b f(x) g(x) dx \right)^2 - 4 \int_a^b f^2(x) dx \int_a^b g^2(x) dx \leq 0$$

$$\text{由 } 1^\circ, 2^\circ \text{ 即有 } \left(\int_a^b f(x) g(x) dx \right)^2 \leq \int_a^b f^2(x) dx \int_a^b g^2(x) dx$$

$$\text{例: 设 } f(x) \in C(a, b), f(x) \geq 0, \text{ 证明: } \int_a^b f(x) dx = 0 \Leftrightarrow f(x) = 0$$

$$\text{证. 必要性显然}$$

充分性: 假设 $f(x) \neq 0$, 则 $\exists x_0 \in (a, b)$, 使 $f(x_0) > 0$,

$f(x) \in C(a, b)$, $\exists \delta > 0$, $(x_0 - \delta, x_0 + \delta) \subset (a, b)$,

$x \in (x_0 - \delta, x_0 + \delta)$, $f(x) > \frac{1}{2}f(x_0)$

$$\int_a^b f(x) dx = \int_a^{x_0-\delta} f(x) dx + \int_{x_0-\delta}^{x_0+\delta} f(x) dx + \int_{x_0+\delta}^b f(x) dx$$

$$\geq \int_{x_0-\delta}^{x_0+\delta} f(x) dx > \frac{1}{2}f(x_0) \cdot 2\delta = f(x_0) \cdot \delta > 0 \text{ 矛盾}$$

假设不成立, 故 $f(x) = 0$

2 微积分基本定理

定义 设 $f(x)$ 在 $[a, b]$ 上可积, $x_0 \in [a, b]$, 则 $\forall x \in [x_0, b]$,

$f(x)$ 在 $[x_0, x]$ 上一定可积, 称 $\int_{x_0}^x f(t) dt$ 为 $f(x)$ 的一个

积分上限函数, 记为 $\varphi(x) = \int_{x_0}^x f(t) dt$.

引理 设 $f(x)$ 在 $[a, b]$ 上可积, 则积分上限函数 $\varphi(x) = \int_a^x f(t) dt$ 在 $[a, b]$ 上连续.

引理 设 $f(x)$ 在 $[a, b]$ 上可积, $x_0 \in (a, b)$, $\lim_{x \rightarrow x_0^+} f(x) = A$, $\lim_{x \rightarrow x_0^-} f(x) = B$, 则积分上限函数 $\varphi(x) = \int_a^x f(x) dx$ 在 x_0 处有右导数 $\varphi'_+(x_0) = A$, 左导数 $\varphi'_-(x_0) = B$.

定理 (微积分基本定理)

1° 设 $f(x)$ 在 $[a, b]$ 上连续, 则积分上限函数 $\varphi(x) = \int_a^x f(t) dt$ 在 $[a, b]$ 上可导, 且 $\varphi'(x) = f(x)$.

2° 设 $f(x)$ 在 $[a, b]$ 上连续, $F(x)$ 为 $f(x)$ 在 $[a, b]$ 上的一个原函数, 则 $\int_a^b f(x) dx = F(b) - F(a) = F(x) \Big|_a^b$ (牛顿-莱布尼兹公式)

推论 设 $f(x)$ 连续, $g(x), h(x)$ 可导,

$$\text{则 } \left(\int_{g(x)}^{h(x)} f(t) dt \right)' = f(h(x)) \cdot h'(x) - f(g(x)) g'(x)$$

例 设 $f(x)$ 连续, $g(x) = \int_x^{x^2} x f(t) dt$, 求 $g'(x)$.

$$g'(x) = x \left(\int_x^{x^2} f(t) dt \right)' = \int_x^{x^2} f(t) dt + x (f(x^2) \cdot 2x - f(x))$$

例 $\lim_{x \rightarrow 0^+} \frac{\int_0^{\sin x} \sqrt{\tan t} dt}{\int_0^{\tan x} \sqrt{\sin t} dt} \stackrel{0}{=} \lim_{x \rightarrow 0^+} \frac{\sqrt{\tan(\sin x)} \cdot \cos x}{\sqrt{\sin(\tan x)} \cdot \sec^2 x}$

$$= \sqrt{\lim_{x \rightarrow 0^+} \frac{\tan(\sin x)}{\sin(\tan x)}} = 1$$

例 求 $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right)$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{1}{1+\frac{1}{n}} + \frac{1}{1+\frac{2}{n}} + \cdots + \frac{1}{1+\frac{n}{n}} \right)$$

$$\text{令 } f(x) = \frac{1}{1+x}, \text{ 上式} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n f\left(\frac{i}{n}\right) = \int_0^1 f(x) dx$$

$$\text{故原式} = \int_0^1 \frac{1}{1+x} dx = \ln(1+x) \Big|_0^1 = \ln 2$$

例 求 $\lim_{n \rightarrow \infty} \left(\frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{n+2n-3} \right)$

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{n+2n-3} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{1}{1+\frac{2}{n}} + \frac{1}{1+\frac{3}{n}} + \cdots + \frac{1}{1+\frac{2n-3}{n}} \right)$$

$$= \int_0^2 \frac{1}{1+x} dx = \ln(1+x) \Big|_0^2 = \ln 3$$

3 定积分的计算

定理 设 $f(x)$ 在 $[a, b]$ 上连续, 对于 $x = \varphi(t)$, 有 α, β 满足

(1) $\varphi(\alpha) = a, \varphi(\beta) = b$;

(2) 在 α, β 界定的区间上, $a \leq \varphi(t) \leq b$;

(3) 在 α, β 界定的区间上, $\varphi(t)$ 有连续导数.

$$\text{则 } \int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt.$$

例 设 $f(x)$ 在 $[0, 1]$ 上连续, 证明:

(1) $\int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx$

(2) $\int_0^{\pi} f(\sin x) dx = 2 \int_0^{\frac{\pi}{2}} f(\sin x) dx$

(3) $\int_0^{\pi} x f(\sin x) dx = \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx$

(4) 求 $\int_0^{\pi} \frac{x \sin x}{1 + \cos x} dx$

证. (1) $\int_0^{\frac{\pi}{2}} f(\sin x) dx \xrightarrow{x = \frac{\pi}{2} - t} \int_{\frac{\pi}{2}}^0 f(\sin(\frac{\pi}{2} - t)) (-1) dt$
 $= \int_0^{\frac{\pi}{2}} f(\cos t) dt = \int_0^{\frac{\pi}{2}} f(\cos x) dx$

(2) $\int_0^{\pi} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\sin x) dx + \int_{\frac{\pi}{2}}^{\pi} f(\sin x) dx$
 $\xrightarrow{x = \pi - t} \int_0^{\frac{\pi}{2}} f(\sin x) dx + \int_0^{\frac{\pi}{2}} f(\sin(\pi - t)) dt$
 $= 2 \int_0^{\frac{\pi}{2}} f(\sin x) dx$

$$\begin{aligned}
 (3) \quad \int_0^{\pi} x f(\sin x) dx & \stackrel{x=\pi-t}{=} \int_{\pi}^0 (\pi-t) f(\sin(\pi-t)) d(\pi-t) \\
 & = \int_0^{\pi} (\pi-x) f(\sin x) dx = \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx \\
 \text{故 } \int_0^{\pi} x f(\sin x) dx & = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx \stackrel{\text{由(2)}}{=} \pi \int_0^{\frac{\pi}{2}} f(\sin x) dx
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx & = \int_0^{\pi} \frac{x \sin x}{2-\sin^2 x} dx \stackrel{\text{由(3)}}{=} \pi \int_0^{\frac{\pi}{2}} \frac{\sin x}{2-\sin^2 x} dx \\
 & = -\pi \int_0^{\frac{\pi}{2}} \frac{1}{1+\cos^2 x} d\cos x = -\pi \arctan \cos x \Big|_0^{\frac{\pi}{2}} = \frac{\pi^2}{4}
 \end{aligned}$$

例1: 求 $\int_{-1}^1 \left(\frac{\cos^2 x}{1+e^x} + \frac{\sin^2 x}{1+e^{x \sin x}} \right) dx$

$f(x) = \frac{\sin^2 x}{1+e^{x \sin x}}$ 为奇函数, 从而 $\int_{-1}^1 f(x) dx = 0$.

$$\begin{aligned}
 \text{故原式} & = \int_{-1}^1 \frac{\cos^2 x}{1+e^x} dx = \int_{-1}^0 \frac{\cos^2 x}{1+e^x} dx + \int_0^1 \frac{\cos^2 x}{1+e^x} dx \\
 & = \int_0^1 \frac{\cos^2(-x)}{1+e^{-x}} dx + \int_0^1 \frac{\cos^2 x}{1+e^x} dx \\
 & = \int_0^1 \cos^2 x dx = \int_0^1 \frac{1+\cos 2x}{2} dx \\
 & = \left(\frac{1}{2}x + \frac{1}{4}\sin 2x \right) \Big|_0^1 = \frac{1}{2} + \frac{1}{4}\sin 2
 \end{aligned}$$

例2: 设 $f(x)$ 连续, $f(x) > 0$, 求 $\int_1^3 \frac{f(x)}{f(x)+f(4-x)} dx$.

$$\begin{aligned}
 \int_1^3 \frac{f(x)}{f(x)+f(4-x)} dx & \stackrel{t=4-x}{=} \int_3^1 \frac{f(4-t)}{f(t)+f(4-t)} (-1) dt = \int_1^3 \frac{f(4-x)}{f(x)+f(4-x)} dx \\
 & = \frac{1}{2} \int_1^3 \left[\frac{f(x)}{f(x)+f(4-x)} + \frac{f(4-x)}{f(x)+f(4-x)} \right] dx = \frac{1}{2} \int_1^3 1 dx = 1
 \end{aligned}$$

定理 设 $f(x)$, $g(x)$ 在 $[a, b]$ 上有连续导数,

$$\text{则 } \int_a^b f'(x)g(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f(x)g'(x)dx.$$

例: $\int_0^1 x \arctan x dx = \frac{1}{2} x^2 \arctan x \Big|_0^1 - \int_0^1 \frac{x^2}{2} \cdot \frac{1}{1+x^2} dx$
 $= \frac{\pi}{8} - \frac{1}{2} \int_0^1 (1 - \frac{1}{1+x^2}) dx = \frac{\pi}{8} - \frac{1}{2} (x - \arctan x) \Big|_0^1$
 $= \frac{\pi}{8} - \frac{1}{2}$

例: 设 $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$, 求 I_n 的递推关系式.

$$I_0 = \frac{\pi}{2}, I_1 = 1, n \geq 2 \text{ 时},$$

$$\begin{aligned} I_n &= \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \sin^{n-1} x d(-\cos x) \\ &= -\cos x \sin^{n-1} x \Big|_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x (1 - \sin^2 x) dx \\ &= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx \end{aligned}$$

$$\text{即 } I_n = (n-1)I_{n-2} - (n-1)I_n, \quad I_n = \frac{n-1}{n} I_{n-2}.$$

例: 求 $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n^2} \ln(1 + \frac{i^2}{n^2})$

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n^2} \ln(1 + \frac{i^2}{n^2}) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{1}{n} \ln(1 + \frac{i^2}{n^2}) \\ &= \int_0^1 x \ln(1+x^2) dx \stackrel{t=x^2}{=} \int_0^1 \frac{1}{2} \ln(1+t) dt = \int_0^1 \frac{1}{2} \ln(1+t) d(1+t) \\ &= \frac{1}{2} (1+t) \ln(1+t) \Big|_0^1 - \frac{1}{2} \int_0^1 \frac{1}{1+t} (t+1) dt = \ln 2 - \frac{1}{2} \end{aligned}$$

例: 求 $P: \begin{cases} x = t - \sin t \\ y = 1 - \cos t \end{cases} (0 \leq t \leq 2\pi)$, 求 P 与 x 轴所围的面积 D .

$$\begin{aligned} D &= \int_0^{2\pi} f(x) dx = \int_0^{2\pi} (1 - \cos t) d(t - \sin t) = \int_0^{2\pi} (1 - \cos t)^2 dt \\ &= \int_0^{2\pi} (2\sin^2 \frac{t}{2})^2 dt = 4 \int_0^{2\pi} \sin^4 \frac{t}{2} dt = 8 \int_0^{\pi} \sin^4 u du \\ &= 16 \int_0^{\frac{\pi}{2}} \sin^4 u du = 16 I_4 = 16 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = 3\pi \end{aligned}$$

例: 设 $f(x)$ 连续, $f(x) = x^2 + \int_0^1 x f(x) dx$, 求 $f(x)$.

$$\text{设 } \int_0^1 x f(x) dx = A, \quad f(x) = x^2 + A$$

$$\int_0^1 x f(x) dx = \int_0^1 (x^3 + Ax) dx = \left(\frac{1}{4} x^4 + \frac{1}{2} A x^2 \right) \Big|_0^1 = \frac{1}{4} + \frac{1}{2} A = A$$

$$A = \frac{1}{2}, \text{ 故 } f(x) = x^2 + \frac{1}{2}.$$

4 反常积分

定义 设任取 $b > a$, $f(x)$ 在 $[a, b]$ 均可积, 称 $\lim_{b \rightarrow +\infty} \int_a^b f(x) dx$ 为 $f(x)$ 在 $[a, +\infty)$ 上的反常积分 (广义积分) 记为 $\int_a^{+\infty} f(x) dx$. 当该极限存在时, 称反常积分 $\int_a^{+\infty} f(x) dx$ 收敛, 否则称其发散.

类似地, 可定义 $\int_{-\infty}^0 f(x) dx$ 和 $\int_{-\infty}^{+\infty} f(x) dx$.

例 讨论广义积分 $\int_a^{+\infty} \frac{1}{(x+b)^p} dx$, $a > -b$ 的敛散性.

$$\int_a^{+\infty} \frac{1}{(x+b)^p} dx = \begin{cases} \frac{(x+b)^{1-p}}{1-p} \Big|_a^{+\infty} = \begin{cases} +\infty, & p < 1, \text{ 发散} \\ \frac{(a+b)^{1-p}}{p-1}, & p > 1, \text{ 收敛} \end{cases} \\ \ln(x+b) \Big|_a^{+\infty} = +\infty, & p = 1, \text{ 发散} \end{cases}$$

定理 设 $0 \leq f(x) \leq g(x)$, $x \in [a, +\infty)$, 若 $\forall b \in (a, +\infty)$, $f(x)$, $g(x)$ 在 $[a, b]$ 上可积, 则

(1) 当 $\int_a^{+\infty} g(x) dx$ 收敛时, $\int_a^{+\infty} f(x) dx$ 收敛,

(2) 当 $\int_a^{+\infty} f(x) dx$ 发散时, $\int_a^{+\infty} g(x) dx$ 发散.

定理 设 $f(x)$, $g(x)$ 在 $[a, +\infty)$ 非负, 且 $\forall b \in (a, +\infty)$, $f(x)$, $g(x)$ 在 $[a, b]$ 上可积, $\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = A$.

(1) 若 $A = 0$ 且 $\int_a^{+\infty} g(x) dx$ 收敛, 则 $\int_a^{+\infty} f(x) dx$ 收敛.

(2) 若 $A \in (0, +\infty)$, 则 $\int_a^{+\infty} g(x) dx$ 与 $\int_a^{+\infty} f(x) dx$ 的敛散性相同.

(3) 若 $A = +\infty$ 且 $\int_a^{+\infty} g(x) dx$ 发散, 则 $\int_a^{+\infty} f(x) dx$ 发散.

定理 设对 $\forall b \in (a, +\infty)$, $f(x)$ 在 $[a, b]$ 上可积, 若 $\int_a^{+\infty} |f(x)| dx$ 收敛, 则 $\int_a^{+\infty} f(x) dx$ 收敛, 且 $|\int_a^{+\infty} f(x) dx| \leq \int_a^{+\infty} |f(x)| dx$.

例 判断敛散性

(1) $\int_1^{+\infty} \frac{x^2}{x^4+x+1} dx$

$$\lim_{x \rightarrow +\infty} \frac{\frac{x^2}{x^4+x+1}}{\frac{1}{x^2}} = \lim_{x \rightarrow +\infty} \frac{x^4}{x^4+x+1} = 1, \text{ 且 } x > 1 \text{ 时, } \frac{1}{x^2} > 0, \frac{x^2}{x^4+x+1} > 0,$$

$$\text{而 } \int_1^{+\infty} \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^{+\infty} = -1 \text{ 收敛, 故 } \int_1^{+\infty} \frac{x^2}{x^4+x+1} dx \text{ 收敛.}$$

(2) $\int_2^{+\infty} \frac{1}{x \ln^2 x (2 + \sin x)} dx$

$$0 < \frac{1}{x \ln^2 x (2 + \sin x)} < \frac{1}{x \ln^2 x}, \quad \int_2^{+\infty} \frac{1}{x \ln^2 x} dx = -\frac{1}{\ln x} \Big|_2^{+\infty} = \frac{1}{\ln 2} \text{ 收敛}$$

$$\text{故 } \int_2^{+\infty} \frac{1}{x \ln^2 x (2 + \sin x)} dx \text{ 收敛}$$

定义 设 $\forall t \in [a, b)$, $f(x)$ 在 $[a, t]$ 上有界, $x \rightarrow b^-$ 时, $f(x)$ 无界, 则称 b 为 $f(x)$ 的一个瑕点.

定义 设 b 为 $f(x)$ 的一个瑕点, 对 $\forall t \in [a, b)$, $f(x)$ 在 $[a, t]$ 可积, 称 $\lim_{t \rightarrow b^-} \int_a^t f(x) dx$ 为 $f(x)$ 的反常积分 (瑕积分), 记为 $\int_a^b f(x) dx$, 当该极限存在时, 称反常积分 $\int_a^b f(x) dx$ 收敛, 否则称其收敛.

例 讨论 $\int_a^b \frac{1}{(x-a)^p} dx$ 的敛散性.

$$\int_a^b \frac{1}{(x-a)^p} dx = \begin{cases} \ln|x-a| \Big|_a^b = +\infty, & p=1, \text{ 发散} \\ \frac{(x-a)^{1-p}}{1-p} \Big|_a^b = \begin{cases} -\infty, & p>1, \text{ 发散} \\ \frac{(b-a)^{1-p}}{1-p}, & p<1, \text{ 收敛.} \end{cases} \end{cases}$$

例 设 $\int_0^{+\infty} \frac{\ln^b(1+x)}{x^p(1+x)^a} dx$ 收敛, $b>0$, 求 a, b, p 的关系.

$$\int_0^{+\infty} \frac{\ln^b(1+x)}{x^p(1+x)^a} dx = \int_0^2 \frac{\ln^b(1+x)}{x^p(1+x)^a} dx + \int_2^{+\infty} \frac{\ln^b(1+x)}{x^p(1+x)^a} dx$$

$$\frac{\ln^b(1+x)}{x^p(1+x)^a} > 0,$$

$$\lim_{x \rightarrow 0} \frac{\frac{\ln^b(1+x)}{x^p(1+x)^a}}{\frac{1}{x^{p-b}}} = 1, \quad \int_0^2 \frac{1}{x^{p-b}} dx \text{ 收敛. } p-b < 1$$

$$\lim_{x \rightarrow +\infty} \frac{\frac{\ln^b(1+x)}{x^p(1+x)^a}}{\frac{\ln^b x}{x^{a+p}}} = 1, \quad \int_2^{+\infty} \frac{\ln^b x}{x^{a+p}} dx \text{ 收敛, } a+p > 1$$

$$\text{故 } \begin{cases} p-b < 1 \\ a+p > 1 \end{cases}$$

例: (1) $\int_0^{\pi} \frac{1}{1+3\sin^2 x} dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+3\sin^2 x} dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1}{1+3\sin^2 x} dx$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1+(2\tan x)^2} dx + \int_{\frac{\pi}{2}}^{\pi} \frac{1}{1+(2\tan x)^2} dx$$

$$= \frac{1}{2} \arctan(2\tan x) \Big|_0^{\frac{\pi}{2}-} + \frac{1}{2} \arctan(2\tan x) \Big|_{\frac{\pi}{2}+}^{\pi}$$

$$= \frac{\pi}{4} - (-\frac{\pi}{4}) = \frac{\pi}{2}$$

(2) $\int_0^{+\infty} \frac{1}{(1+x^2)(1+x^a)} dx \quad (a>0) = \int_0^1 \frac{1}{(1+x^2)(1+x^a)} dx + \int_1^{+\infty} \frac{1}{(1+x^2)(1+x^a)} dx$

$$\stackrel{x=\frac{1}{t}}{=} \int_0^1 \frac{1}{(1+x^2)(1+x^a)} dx + \int_1^0 \frac{1}{(1+\frac{1}{t^2})(1+\frac{1}{t^a})} d\frac{1}{t}$$

$$= \int_0^1 \frac{1}{(1+x^2)(1+x^a)} dx + \int_0^1 \frac{x^a}{(1+x^2)(1+x^a)} dx$$

$$= \int_0^1 \frac{1}{1+x^2} dx = \arctan x \Big|_0^1 = \frac{\pi}{4}$$

(3) $\int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx \stackrel{x=\frac{\pi}{4}-t}{=} \int_{\frac{\pi}{4}}^0 \ln[1+\tan(\frac{\pi}{4}-t)] d(\frac{\pi}{4}-t)$

$$= \int_0^{\frac{\pi}{4}} \ln(1 + \frac{1-\tan x}{1+\tan x}) dx = \int_0^{\frac{\pi}{4}} \ln \frac{2}{1+\tan x} dx$$

$$= \int_0^{\frac{\pi}{4}} \ln 2 dx - \int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx$$

即 $\int_0^{\frac{\pi}{4}} \ln(1+\tan x) dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} \ln 2 dx = \frac{1}{2} \ln 2 \times \frac{\pi}{4} = \frac{\pi \ln 2}{8}$

(4) $\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^{\sqrt{2}}} dx \stackrel{x=\frac{\pi}{2}-t}{=} \int_{\frac{\pi}{2}}^0 \frac{1}{1+(\tan(\frac{\pi}{2}-t))^{\sqrt{2}}} d(\frac{\pi}{2}-t)$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1+(\cot x)^{\sqrt{2}}} dx = \int_0^{\frac{\pi}{2}} \frac{(\tan x)^{\sqrt{2}}}{1+(\tan x)^{\sqrt{2}}} dx$$

从而 $\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^{\sqrt{2}}} dx = \frac{1}{2} (\int_0^{\frac{\pi}{2}} \frac{1}{1+(\tan x)^{\sqrt{2}}} dx + \int_0^{\frac{\pi}{2}} \frac{(\tan x)^{\sqrt{2}}}{1+(\tan x)^{\sqrt{2}}} dx)$

$$= \int_0^{\frac{\pi}{2}} 1 dx = x \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

例: 设 $f(u) = 1$, $x \geq 1$ 时, $f'(x) = \frac{1}{x^2 + f^2(x)}$, 证明: $\lim_{x \rightarrow +\infty} f(x)$ 存在.

证: $f'(x)$ 在 $[1, +\infty)$ 上连续, $\int_1^t f'(x) dx = f(t) - f(1)$

$f'(x) > 0$, $f(x)$ 在 $[1, +\infty)$ 上 $\uparrow$, $f(x) \geq f(1) = 1$,

$$0 < f'(x) = \frac{1}{x^2 + f^2(x)} \leq \frac{1}{1 + x^2},$$

$$\text{从而 } \int_1^{+\infty} f'(x) dx \leq \int_1^{+\infty} \frac{1}{1+x^2} dx = \arctan x \Big|_1^{+\infty} = \frac{\pi}{4}$$

$$\text{有 } \int_1^t f'(x) dx \leq \int_1^{+\infty} f'(x) dx \leq \frac{\pi}{4}, \quad f(t) \leq \frac{\pi}{4} + 1, \quad \forall t \in [1, +\infty)$$

故 $\lim_{x \rightarrow +\infty} f(x)$ 存在.

例: 求 $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\ln(1 + \frac{k}{n})}{n + \frac{k}{n}}$.

$$\frac{\ln(1 + \frac{k}{n})}{n+1} \leq \frac{\ln(1 + \frac{k}{n})}{n + \frac{k}{n}} \leq \frac{\ln(1 + \frac{k}{n})}{n}$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \ln(1 + \frac{k}{n}) = \int_0^1 \ln(1+x) dx = (x+1)(\ln(1+x) - 1) \Big|_0^1 = 2\ln 2 - 1$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+1} \ln(1 + \frac{k}{n}) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n+1} \cdot \frac{1}{n} \ln(1 + \frac{k}{n}) = \lim_{n \rightarrow \infty} \frac{n}{n+1} \cdot \int_0^1 \ln(1+x) dx \\ &= 2\ln 2 - 1 \end{aligned}$$

$$\text{故 } \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\ln(1 + \frac{k}{n})}{n + \frac{k}{n}} = 2\ln 2 - 1$$

例: 设 $f(x) \in C'[0, 1]$, $0 \leq f'(x) \leq 1$, $f(0) = 0$.

$$\text{证明: } \left(\int_0^1 f(x) dx \right)^2 \geq \int_0^1 f^3(x) dx$$

证. 令 $g(t) = (\int_0^t f(x) dx)^2 - \int_0^t f^2(x) dx$.

$$g'(t) = 2 \int_0^t f(x) dx \cdot f(t) - f^2(t) = f(t) [2 \int_0^t f(x) dx - f^2(t)]$$

$$\text{令 } h(t) = 2 \int_0^t f(x) dx - f^2(t), \quad h'(t) = 2f(t)[1 - f'(t)]$$

$x \in [0, 1], f'(x) \geq 0$, 故 $f(x) \geq f(0) = 0$

从而 $h(t) \geq 0, g'(t) \geq 0, g(t) \geq g(0) = 0$.

$g(1) \geq 0$, 即 $(\int_0^1 f(x) dx)^2 \geq \int_0^1 f^2(x) dx$

例: 设 $f(x) \in C[0, 1], 0 < m \leq f(x) \leq M, x \in [0, 1]$.

$$\text{证明: } 1 \leq \int_0^1 \frac{1}{f(x)} dx \cdot \int_0^1 f(x) dx \leq \frac{(M+m)^2}{4Mm}$$

$$\text{证. } \int_0^1 \frac{1}{f(x)} dx \int_0^1 f(x) dx = \int_0^1 \left(\frac{1}{\sqrt{f(x)}} \right)^2 dx \cdot \int_0^1 (\sqrt{f(x)})^2 dx$$

$$\geq \left(\int_0^1 \frac{1}{\sqrt{f(x)}} \cdot \sqrt{f(x)} dx \right)^2 = 1$$

$$0 < m \leq f(x) \leq M, \quad \frac{1}{M} \leq \frac{1}{f(x)} \leq \frac{1}{m}, \quad [f(x) - M][\frac{1}{f(x)} - \frac{1}{m}] \geq 0$$

$$\text{即 } \frac{f(x)}{m} + \frac{M}{f(x)} \leq 1 + \frac{M}{m}, \quad \frac{1}{m} \int_0^1 f(x) dx + M \int_0^1 \frac{1}{f(x)} dx \leq 1 + \frac{M}{m}$$

$$\frac{1}{m} \int_0^1 f(x) dx + M \int_0^1 \frac{1}{f(x)} dx \geq 2 \sqrt{\frac{M}{m}} \sqrt{\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx}$$

$$\text{故 } 2 \sqrt{\frac{M}{m}} \sqrt{\int_0^1 f(x) dx \int_0^1 \frac{1}{f(x)} dx} \geq 1 + \frac{M}{m}$$

$$\text{即 } \int_0^1 \frac{1}{f(x)} dx \cdot \int_0^1 f(x) dx \leq \frac{(M+m)^2}{4Mm}$$

例: 设 $f(x) \in C[0, 1]$, 且 $f(x) > 0, x \in [0, 1]$.

$$\text{证明: } \int_0^1 \ln f(x) dx \leq \ln \int_0^1 f(x) dx$$

证. (1) 令 $g(x) = \ln x$, $g'(x) = \frac{1}{x}$, $g''(x) = -\frac{1}{x^2}$, 记 $t_0 = \int_0^1 f(x) dx$, $f(x) > 0$, $t_0 > 0$.

$$g(x) = g(t_0) + g'(x)(x - t_0) + \frac{1}{2} g''(\xi)(x - t_0)^2 \leq g(t_0) + g'(t_0)(x - t_0)$$

$$\text{即 } \ln x \leq \ln t_0 + \frac{x - t_0}{t_0}, \text{ 故 } \ln f(x) \leq \ln t_0 + \frac{f(x) - t_0}{t_0}$$

$$\int_0^1 \ln f(x) dx \leq \int_0^1 \ln t_0 dx + \frac{\int_0^1 f(x) dx - \int_0^1 t_0 dx}{\int_0^1 t_0 dx}$$

$$= \ln t_0 + \frac{t_0 - t_0}{t_0} = \ln \int_0^1 f(x) dx$$

(2) $y = \ln x$ 为凸函数.

$$\int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n} [f(\frac{1}{n}) + f(\frac{2}{n}) + \dots + f(\frac{n}{n})]$$

$$\ln \int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \ln \frac{1}{n} [f(\frac{1}{n}) + f(\frac{2}{n}) + \dots + f(\frac{n}{n})]$$

$$\geq \lim_{n \rightarrow \infty} \frac{1}{n} [\ln f(\frac{1}{n}) + \ln f(\frac{2}{n}) + \dots + \ln f(\frac{n}{n})]$$

$$= \int_0^1 \ln f(x) dx$$

例: 设 $f(x)$ 在 $[a, b]$ 上有二阶导数,

$$\text{证明: } f''(x) \geq 0, x \in [a, b] \Leftrightarrow \int_a^b f(x) dx \geq (b-a)f(\frac{a+b}{2}), \forall a, b \in \mathbb{R}$$

证. 充分性:

$$\text{令 } g(t) = \int_a^t f(x) dx,$$

$$g'(t) = f(t), g''(t) = f'(t), g'''(t) = f''(t) \geq 0$$

$$g(t) = g(\frac{a+b}{2}) + g'(\frac{a+b}{2})(t - \frac{a+b}{2}) + \frac{1}{2} g''(\frac{a+b}{2})(t - \frac{a+b}{2})^2 + \frac{1}{6} g'''(\xi)(t - \frac{a+b}{2})^3, \xi \text{ 介于 } t \text{ 与 } \frac{a+b}{2} \text{ 之间.}$$

$$= g(\frac{a+b}{2}) + f(\frac{a+b}{2})(t - \frac{a+b}{2}) + \frac{1}{2} f'(\frac{a+b}{2})(t - \frac{a+b}{2})^2 + \frac{1}{6} f'(\xi)(t - \frac{a+b}{2})^3$$

$$\textcircled{1} \quad 0 = g(a) = g\left(\frac{a+b}{2}\right) + f\left(\frac{a+b}{2}\right) \frac{a-b}{2} + \frac{1}{2} f'\left(\frac{a+b}{2}\right) \left(\frac{a-b}{2}\right)^2 + \frac{1}{6} f''(\xi_1) \left(\frac{a-b}{2}\right)^3$$

$$\textcircled{2} \quad \int_a^b f(x) dx = g\left(\frac{a+b}{2}\right) + f\left(\frac{a+b}{2}\right) \left(-\frac{a-b}{2}\right) + \frac{1}{2} f'\left(\frac{a+b}{2}\right) \left(\frac{a-b}{2}\right)^2 + \frac{1}{6} f''(\xi_2) \left(-\frac{a-b}{2}\right)^3$$

$$\textcircled{2} - \textcircled{1} \text{ 得 } \int_a^b f(x) dx = (b-a) f\left(\frac{a+b}{2}\right) + \frac{f''(\xi_1) + f''(\xi_2)}{3} \left(\frac{b-a}{2}\right)^3$$

由于 $f''(x) \geq 0$, 故 $\int_a^b f(x) dx \geq (b-a) f\left(\frac{a+b}{2}\right)$.

必要性: 取积分上下限为 $x_0 - \delta$ 和 $x_0 + \delta$, $x_0 \in \mathbb{R}$, $\delta > 0$.

$$\text{则 } 0 \leq \lim_{\delta \rightarrow 0^+} \frac{\int_{x_0-\delta}^{x_0+\delta} f(x) dx - 2\delta f(x_0)}{[x_0+\delta - (x_0-\delta)]^3} = \lim_{\delta \rightarrow 0^+} \frac{\int_{x_0-\delta}^{x_0+\delta} f(x) dx - 2\delta f(x_0)}{8\delta^3}$$

$$\stackrel{0}{=} \lim_{\delta \rightarrow 0^+} \frac{f(x_0+\delta) + f(x_0-\delta) - 2f(x_0)}{24\delta^2}$$

$$= \lim_{\delta \rightarrow 0^+} \frac{[f(x_0) + f'(x_0)\delta + \frac{1}{2}f''(x_0)\delta^2 + o(\delta^2)] + [f(x_0) - f'(x_0)\delta + \frac{1}{2}f''(x_0)\delta^2 + o(\delta^2)] - 2f(x_0)}{24\delta^2}$$

$$= \lim_{\delta \rightarrow 0^+} \frac{f''(x_0)\delta^2 + o(\delta^2)}{24\delta^2} = \frac{1}{24} f''(x_0), \text{ 故 } f''(x_0) \geq 0$$

综上, 命题得证.

例: 设 $f(x) \in C^1[a, b]$, $f(a) = f(b) = 0$.

$$\text{证明: } \left| \int_a^b f(x) dx \right| \leq \frac{(b-a)^2}{4} \max_{a \leq x \leq b} |f'(x)|$$

证. (一) 令 $g(t) = \int_a^t f(x) dx$, $g'(t) = f(t)$, $g''(t) = f'(t)$

$$h(t) = \int_t^b f(x) dx, \quad h'(t) = -f(t), \quad h''(t) = -f'(t)$$

$$g(t) = g(a) + g'(a)(t-a) + \frac{1}{2} g''(\xi_1)(t-a)^2$$

$$= \frac{1}{2} f'(\xi_1)(t-a)^2, \quad \xi_1 \in (a, t)$$

$$h(t) = h(b) + h'(b)(t-b) + \frac{1}{2} h''(\xi_2)(t-b)^2$$

$$= \frac{1}{2} f'(\xi_2)(t-b)^2, \quad \xi_2 \in (t, b)$$

$$\left| \int_a^b f(x) dx \right| = \left| g\left(\frac{a+b}{2}\right) + h\left(\frac{a+b}{2}\right) \right| = \left| \frac{1}{2} f'(\xi_1) \left(\frac{b-a}{2}\right)^2 + \frac{1}{2} f'(\xi_2) \left(\frac{b-a}{2}\right)^2 \right|$$

$$\leq \frac{|f'(\xi_1)| + |f'(\xi_2)|}{2} \cdot \frac{(b-a)^2}{4} \leq \frac{(b-a)^2}{4} \max_{a \leq x \leq b} |f'(x)|$$

(二) $f(x) \in C^1[a, b]$, 设 $M = \max_{a \leq x \leq b} |f'(x)|$

$$f(x) = f'(\xi_1)(x-a), \quad \xi_1 \in (a, x)$$

$$f(x) = f'(\xi_2)(x-b), \quad \xi_2 \in (x, b)$$

$$\left| \int_a^b f(x) dx \right| = \left| \int_a^{\frac{a+b}{2}} f(x) dx + \int_{\frac{a+b}{2}}^b f(x) dx \right|$$

$$= \left| \int_a^{\frac{a+b}{2}} f'(\xi_1)(x-a) dx + \int_{\frac{a+b}{2}}^b f'(\xi_2)(b-x) dx \right|$$

$$\leq \int_a^{\frac{a+b}{2}} |f'(\xi_1)|(x-a) dx + \int_{\frac{a+b}{2}}^b |f'(\xi_2)|(b-x) dx$$

$$\leq M \int_a^{\frac{a+b}{2}} (x-a) dx + M \int_{\frac{a+b}{2}}^b (b-x) dx$$

$$= M \cdot \frac{(b-a)^2}{4} = \frac{(b-a)^2}{4} \max_{a \leq x \leq b} |f'(x)|$$

5 定积分的应用

微元法

求 Q , 当 Q 满足

1) Q 对应于一个区间, 且将 $[a, b]$ 分割成 n 个小区间时, Q 相应分割成 $\Delta Q_1, \Delta Q_2, \dots, \Delta Q_n$, 且 $Q = \sum_{i=1}^n \Delta Q_i$ (称 Q 在区间 $[a, b]$)

上有分块可加性).

(2) Q 在典型区间 $[x, x+\Delta x]$ 上对应量 $\Delta Q \approx f(x) \Delta x$, 其中 $f(x)$ 在 $[a, b]$ 上连续.

(3) 从而 Q 的微元 $dQ = f(x) dx$.

$$Q = \int_a^b f(x) dx.$$

平面区域的面积

在区间 $[a, b]$ 上,

$y_2(x) \geq y_1(x)$, $x=a$, $x=b$, 面积微元 $dS = [y_2(x) - y_1(x)] dx$,

从而 $S = \int_a^b [y_2(x) - y_1(x)] dx$. (x -型区域)

$x_2(y) \geq x_1(y)$, $y=c$, $y=d$, 面积微元 $dS = [x_2(y) - x_1(y)] dy$,

从而 $S = \int_c^d [x_2(y) - x_1(y)] dy$. (y -型区域)

在区间 $[\alpha, \beta]$ 上,

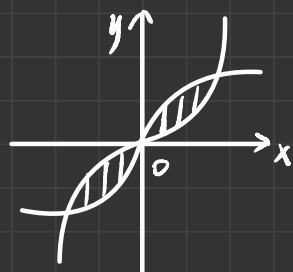
极坐标曲线 $r=r_1(\theta)$, $r=r_2(\theta)$, $r_1 \leq r_2$, $\theta=\alpha$, $\theta=\beta$.

面积微元 $dS = \frac{1}{2} [r_2^2(\theta) - r_1^2(\theta)] d\theta$, 从而 $S = \frac{1}{2} \int_\alpha^\beta [r_2^2(\theta) - r_1^2(\theta)] d\theta$.

例: 求曲线 $y=x^3$ 与 $x=y^3$ 围成图形的面积 D

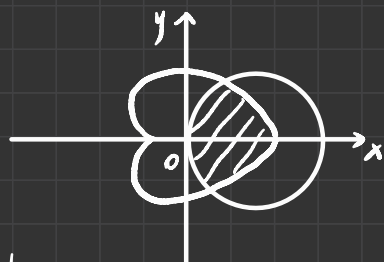
$$\begin{cases} y=x^3 \\ x=y^3 \end{cases} \Rightarrow \text{交点 } (1,1) \text{ 和 } (-1,-1)$$

$$\begin{aligned} D &= 2 \int_0^1 (x^{\frac{1}{3}} - x^3) dx \\ &= 2 \left(\frac{3}{4} x^{\frac{4}{3}} - \frac{1}{4} x^4 \right) \Big|_0^1 = 1 \end{aligned}$$



例1 求极坐标曲线 $r = 3\cos\theta$ 与 $r = 1 + \cos\theta$ 所围图形公共部分面积 D .

$$\begin{cases} r = 3\cos\theta \\ r = 1 + \cos\theta \end{cases} \Rightarrow \text{交点 } (\frac{\pi}{3}, \frac{3}{2}) \text{ 和 } (-\frac{\pi}{3}, -\frac{3}{2})$$



$$\begin{aligned} D &= 2 \left[\frac{1}{2} \int_0^{\frac{\pi}{3}} (1 + \cos\theta)^2 d\theta + \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (3\cos\theta)^2 d\theta \right] \\ &= \int_0^{\frac{\pi}{3}} (1 + 2\cos\theta + \cos^2\theta) d\theta + 9 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos^2\theta d\theta \end{aligned}$$

$$\text{由于 } \int \cos^2\theta d\theta = \int \frac{1 + \cos 2\theta}{2} d\theta = \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta + C$$

$$\begin{aligned} \text{故 } D &= \left(\theta + 2\sin\theta + \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta \right) \Big|_0^{\frac{\pi}{3}} + 9 \cdot \left(\frac{1}{4}\sin 2\theta + \frac{1}{2}\theta \right) \Big|_{\frac{\pi}{3}}^{\frac{\pi}{2}} \\ &= \frac{5\pi}{4} \end{aligned}$$

曲线的弧长

在区间 $[a, b]$ 上,

$y = f(x)$ 连续, 弧长微元 $ds = \sqrt{1 + [f'(x)]^2} dx$, 从而弧长 $s = \int_a^b \sqrt{1 + [f'(x)]^2} dx$

若 $y = f(x)$ 由参数方程

$$\begin{cases} x = x(t) \\ y = y(t) \end{cases}, \alpha \leq t \leq \beta \text{ 确定, 则 } s = \int_{\alpha}^{\beta} \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

在区间 $[\alpha, \beta]$ 上, 极坐标曲线 $r = r(\theta)$,

$$\begin{cases} x = r(\theta) \cos\theta \\ y = r(\theta) \sin\theta \end{cases}$$

$$\text{弧长微元 } ds = \sqrt{(r(\theta) \cos\theta)' ^2 + (r(\theta) \sin\theta)' ^2} d\theta = \sqrt{[r'(\theta)]^2 + r^2(\theta)} d\theta$$

$$\text{从而弧长 } s = \int_{\alpha}^{\beta} \sqrt{[r'(\theta)]^2 + r^2(\theta)} d\theta$$

例: 求曲线 $y=x^2$, $0 \leq x \leq 1$ 的长度 l .

$$l = \int_0^1 \sqrt{1+(y')^2} dx = \int_0^1 \sqrt{1+4x^2} dx = 2 \int_0^1 \sqrt{\frac{1}{4} + x^2} dx$$

$$\text{其中 } \int_0^1 \sqrt{x^2 + \frac{1}{4}} dx = x\sqrt{x^2 + \frac{1}{4}} \Big|_0^1 - \int_0^1 \frac{x^2}{\sqrt{x^2 + \frac{1}{4}}} dx$$

$$= \frac{\sqrt{5}}{2} - \int_0^1 \frac{x^2 + \frac{1}{4} - \frac{1}{4}}{\sqrt{x^2 + \frac{1}{4}}} dx$$

$$= \frac{\sqrt{5}}{2} + \frac{1}{4} \int_0^1 \frac{1}{\sqrt{x^2 + \frac{1}{4}}} dx - \int_0^1 \sqrt{x^2 + \frac{1}{4}} dx$$

$$= \frac{\sqrt{5}}{2} + \frac{1}{4} \ln(x + \sqrt{x^2 + \frac{1}{4}}) \Big|_0^1 - \int_0^1 \sqrt{x^2 + \frac{1}{4}} dx$$

$$= \frac{\sqrt{5}}{2} + \frac{1}{4} \ln(1 + \frac{\sqrt{5}}{2}) - \int_0^1 \sqrt{x^2 + \frac{1}{4}} dx$$

$$\text{从而 } l = 2 \int_0^1 \sqrt{x^2 + \frac{1}{4}} dx = \frac{\sqrt{5}}{2} + \frac{1}{4} \ln(1 + \frac{\sqrt{5}}{2})$$

例: 求曲线 $r=1+\cos\theta$, $0 \leq \theta \leq 2\pi$ 的长度 l .

$$l = \int_0^{2\pi} \sqrt{[r'(\theta)]^2 + r^2(\theta)} d\theta = \int_0^{2\pi} \sqrt{2+2\cos\theta} d\theta = \int_0^{2\pi} \sqrt{4\cos^2 \frac{\theta}{2}} d\theta$$

$$= 2 \int_0^{2\pi} |\cos \frac{\theta}{2}| d\theta = 2 \left(\int_0^{\pi} \cos \frac{\theta}{2} d\theta + \int_{\pi}^{2\pi} -\cos \frac{\theta}{2} d\theta \right)$$

$$= 4 \left(\sin \frac{\theta}{2} \Big|_0^{\pi} - \sin \frac{\theta}{2} \Big|_{\pi}^{2\pi} \right) = 8$$

几何体的体积

设 V 为一几何体, V 对应于一个区间 $[a, b]$.

$x \in [a, b]$, 对应的截面面积 $S(x)$ 在 $[a, b]$ 连续.

则体积微元 $dV = S(x)dx$, 从而 $V = \int_a^b S(x)dx$.

设由 $y=f(x)$, $x=a$, $x=b$ 及 x 轴所围成图形绕 x 轴旋转一周所得旋转体体积为 V .

体积微元 $dv = \pi f^2(x) dx$, 从而 $V = \pi \int_a^b f^2(x) dx$.

设 a, b 同号, $y=f(x)$, $x=a$, $x=b$ 及 x 轴所围成图形绕 y 轴旋转一周所得旋转体体积为 V .

体积微元 $dV = 2\pi x dx \cdot |f(x)|$, 从而 $V = 2\pi \int_a^b x |f(x)| dx$

例: 求 $y=x^3$ 与 $x=1$, x 轴所围图形绕 y 轴旋转一周所得旋转体体积为 V .

$$dV = 2\pi x dx \cdot |f(x)|, \quad V = 2\pi \int_a^b x |f(x)| dx$$

$$V = 2\pi \int_0^1 x \cdot x^3 dx = 2\pi \cdot \frac{1}{5} x^5 \Big|_0^1 = \frac{2\pi}{5}$$

例: 求圆 $(x-4)^2 + (y-3)^2 = 1$ 绕 y 轴旋转一周的体积 V .

$$V = \int_3^5 2\sqrt{1-(4-x)^2} \cdot 2\pi x dx = 4\pi \int_3^5 \sqrt{1-(4-x)^2} x dx$$

$$\stackrel{t=4-x}{=} 4\pi \int_1^{-1} (t+4) \sqrt{1-t^2} dt = 16\pi \int_{-1}^1 \sqrt{1-t^2} dt = 8\pi^2$$

旋转曲面的面积

设 $y=f(x)$, $x \in [a, b]$ 绕 x 轴旋转一周所得旋转曲面的面积为 S ,

面积微元 $dS = 2\pi |f(x)| ds = 2\pi |f(x)| \sqrt{1+[f'(x)]^2} dx$

从而 $S = 2\pi \int_a^b |f(x)| \sqrt{1+[f'(x)]^2} dx$

设 $y=f(x)$, $x \in [a, b]$ 绕 y 轴旋转一周所得旋转曲面的面积为 S ,

$$\text{面积微元 } dS = 2\pi |x| ds = 2\pi |x| \sqrt{1 + [f'(x)]^2} dx,$$

$$\text{从而 } S = 2\pi \int_a^b |x| \sqrt{1 + [f'(x)]^2} dx.$$